Semantic Processing: Experimental and Modeling Evidence from English

Introduction. English sentences with an existential subject and a universal object often exhibit scope ambiguity. For example, *A dog chased every cat* allows both a surface-scope reading (*one dog chased all the cats*) and an inverse-scope reading (*each cat may have been chased by a different dog*), with the latter more difficult to access (Anderson, 2004). While offline scope interpretation has been widely studied (Kurtzman & MacDonald, 1993; Scontras et al., 2017), comparatively fewer studies have examined its online computation to identify the processing costs associated with different interpretations and their time course (cf. Anderson, 2004). Recent work has also highlighted variability in speakers' grammatical representations (Zimmermann et al., 2025) and extended psycholinguistic processing questions to large language models (LLMs; Günther et al., 2025). This study investigates the real-time processing of doubly-quantified English sentences, individual differences related to working memory (WM) and language experience (LE) as indexed by lexical knowledge, and the extent to which surprisal values derived from syntactic structures can predict human semantic processing, as reflected in the location and magnitude of scope-related effects.

Experiments. A non-cumulative word-by-word self-paced reading (SPR) experiment was conducted with 51 English native speakers recruited through Prolific. Twenty-four sets of "a-every" sentences were created, each pairing two context sentences with a continuation sentence that disambiguated toward either a surface- or inverse-scope reading. Another 24 sets served as a baseline to determine whether observed effects reflected genuine scope processing rather than shallow parsing from morphological mismatches. As shown in Table 1, the blue regions mark the critical and spillover Regions of Interest (ROI)s; longer RTs for (1b) than (1a), paired with no differences between (2a) and (2b), were taken as empirical evidence of increased processing costs for inverse scope. A Latin Square ensured that each participant saw 12 items per condition mixed with 48 fillers. Participants also completed a backward digit span test (for WM) and an extended LexTALE (for LE). Surprisal values for each region of the test sentences was derived from GPT-2. Surprisal prediction procedures followed Huang et al. (2024).

Results & Discussion. Human SPR data were trimmed using standard procedures: two participants with filler-comprehension accuracy below 80% were removed, and RTs were trimmed using a 100–5000 ms range, a 2.5 SD cutoff, and log/residual transformations. For surprisal analyses, spillover predictors were created for surprisal, word length, and word frequency for the preceding three words (w–1, w–2, w–3), including frequency × length interactions. Mixed-effects models showed reliably greater processing costs for inverse- than surface-scope in both spillover regions ($ps < .05$; Fig. 1), although a difference in the final spillover region between the baseline conditions ($p = .036$; Fig. 2) leaves open the possibility of shallow processing. LE, but not WM, facilitated inverse scope processing ($p = .007$). Surprisal exhibited strong predictive power, reproducing human-like difficulty patterns for inverse scope relative to surface scope (Fig. 3), and accounted for a substantial proportion of variance (around 30%) (Fig. 4), adding to the research (Huang et al., 2024; Kobzeva & Kush, 2024). Overall, the results demonstrated signs of scope processing effects, reveal meaningful individual differences, and highlight the predictive potential of LLM-derived surprisal to account for human semantic processing.

Table 1. Example sentences across conditions

	Condition	Example sentence	Comprehension question
Scope conditions	Surface	(1a). A dog chased every cat, although the **dog was not** very fast.	How many dogs chased cats? A. Several B. One
	Inverse	(1b). A dog chased every cat, although the **dogs were not** very fast.	
Baseline conditions	Same	(2a). The same dog chased every cat, although the **dog was not** very fast.	
	Different	(2b). A different dog chased every cat, although the **dogs were not** very fast.	

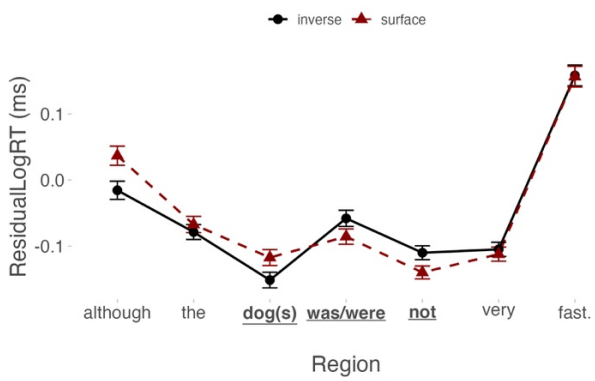

Figure 1. Word-by-word RTs for scope conditions

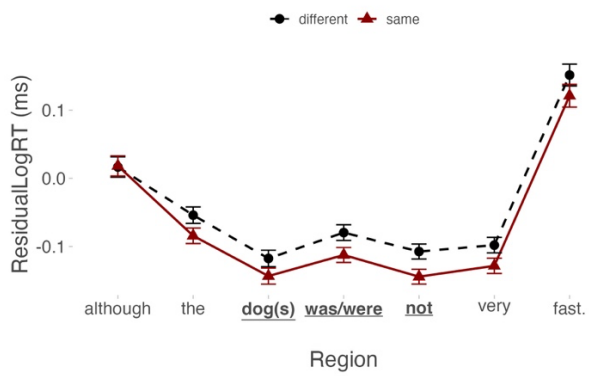

Figure 2. Word-by-word RTs for baseline

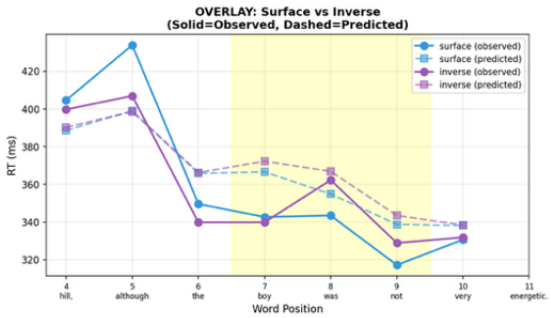

Figure 3. RTs predicted by surprisal

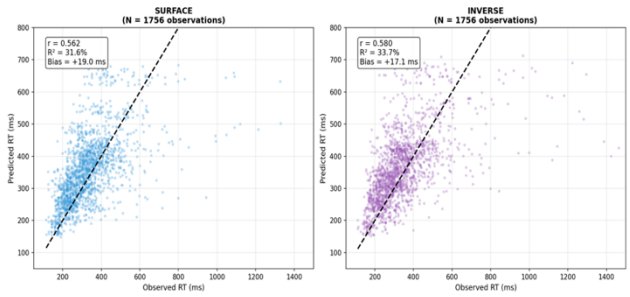

Figure 4. Predicted scope effects magnitude

Selected References:

Anderson, C. (2004). Quantifier scope ambiguity. Northwestern Univ.

Günther, F., Cassani, G., & Günther, F. (2025). LLMs in psycholinguistics.

Kurtzman, H. S., & MacDonald, M. C. (1993). Cognition.

Lemhöfer, K., & Broersma, M. (2012). Behavior Research Methods.

Scontras, G. et al. (2017). Glossa.